

ISE 333 - Quiz 1 & Quiz 2

中英双语试题讲解

Bilingual Questions, Answers, and Explanations

内容范围 / Scope: 按 Quiz1/Quiz2 试卷图片逐题整理；答案优先依据 GUI开发目录中的课程课件，并标注页码或幻灯片编号。

课程 / Course: ISE 333 User Interface Development (Design)

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使用说明 / How to Use This Guide

资料完整性 / Source completeness: Quiz 1 图片中未出现第18、19题，因此本文不臆造题目；Quiz 2 的第10、11、13-16题涉及评估与辛普森悖论，但目录中未提供对应的第5/6章课件，相关条目已逐题标明。

整理原则 / Method: 保留英文题意，给出中文翻译；先给答案，再解释排除逻辑、知识点和课件来源。题目原文存在少量语法问题，讲解中按其明确含义处理。

复习建议 / Study advice:

先用答案速查自测，再遮住答案逐题作答；错题应回到标注课件页复习概念、分类边界和应用条件。

答案速查 / Quick Answer Key

Quiz 1 答案 / Answers	Quiz 2 答案 / Answers
1 A	1 D
2 A	2 C
3 B	3 D
4 B	4 C
5 C	5 D
6 C	6 B
7 D	7 C
8 D	8 B
9 用户错误率	9 C
10 A	10 B
11 T	11 C
12 F	12 A
13 T	13 C
14 F	14 C
15 F	15 C
16 F	16 分层反转
17 T	
18-19 缺图	
20 儿童B/成人A	
21 C	

Quiz 1 逐题讲解 / Detailed Solutions

覆盖 / Coverage: Chapter 1, Chapter 2, Chapter 3 (Guidelines).

Quiz 1 - Q1 (3 pts)

English: The acronym UID in the course context cannot be which one among the following terms?

中文: 在本课程语境中，缩写 UID 不可能表示下列哪一项？

A	United interaction domain	联合交互域
B	User interface development	用户界面开发
C	User interface design	用户界面设计
D	User interface designing	用户界面设计活动

答案 / Answer: A

中文讲解: 课程将 UID 用作 User Interface Development/Design 的缩写。A 与课程术语无关。

English explanation: In this course, UID refers to user interface development/design. Option A is unrelated to the course terminology.

知识点 / Key point: 先识别课程内的专用缩写，不要按一般英语随意扩展首字母。 / Interpret an acronym within the course vocabulary rather than inventing a generic expansion.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.18 and p.38

Quiz 1 - Q2 (3 pts)

English: UID is important not only in terms of technology, but also in terms of its ____.

中文: UID 不仅在技术层面重要，还在什么层面重要？

A	Social impact	社会影响
B	Outcome beauty	结果美观
C	Implementation ease	实现便利
D	Economical guarantee	经济保证

答案 / Answer: A

中文讲解: 课件明确区分个体层面与社会层面的影响：医疗、飞行、学习，以及集体行动、政策参与和治理。

English explanation: The slides explicitly discuss impact at both individual and societal levels, including healthcare, learning, collective action, and policy engagement.

知识点 / Key point: 优秀 UI 的价值不能只用技术完成度衡量，还要看其对个人与社会产生的后果。 / A UI should be evaluated not only by technical completion but also by its consequences for individuals and society.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.18-19

Quiz 1 - Q3 (3 pts)

English: Which statement about UIDers is wrong?

中文: 关于 UID 从业者的陈述，哪一项错误？

A	UIDers are heroes of a profound transformation.	UID 从业者推动了深刻变革。
B	UIDers are more important than all other hardware and software developers.	UID 从业者比所有软硬件开发者都更重要。
C	Their work helped turn personal computers into successful mobile devices.	他们的工作推动个人计算向成功的移动设备演进。
D	Professional desktop applications increasingly gave way to global social tools.	专业桌面应用日益让位于面向全球社区的社会化工具。

答案 / Answer: B

中文讲解: 课件赞扬 UID 的贡献，但没有建立职业高低等级。UI、硬件、软件和其他工程角色相互依赖，因此“更重要”是无依据的绝对化判断。

English explanation: The slides praise the contribution of UID without ranking professions. UI, hardware, software, and other engineering roles are interdependent, so the absolute claim in B is unsupported.

知识点 / Key point: 警惕将“贡献显著”偷换成“比其他角色更重要”。 / Do not turn a claim of major contribution into an unsupported ranking of professions.

课件依据 / **Source:** chap01_Usability of interactive systems.pdf, p.13-14

Quiz 1 - Q4 (3 pts)

English: Which statement is correct?

中文: 下列哪一项正确？

A	HCI does not involve experimental psychology.	HCI 不涉及实验心理学。
B	Life-changing shifts were possible because researchers and UIDers harnessed technology to serve human needs.	研究者和 UID 从业者让技术服务于人的需求，从而促成改变生活的转变。
C	Human-factors and ergonomics experts never contributed to UID.	人因与工效学专家从未参与 UID。
D	UID knowledge is purely management-style knowledge.	UID 知识纯粹属于管理知识。

答案 / Answer: B

中文讲解: HCI

将实验心理学方法应用于计算机科学，并吸收教育心理、图形设计、人因工程、人类学和社会学等贡献，所以 A、C、D 均错误。

English explanation: HCI applies experimental-psychology methods to computing and integrates contributions from psychology, design, ergonomics, anthropology, and sociology. Therefore A, C, and D are false.

知识点 / Key point: HCI/UID 是跨学科工程实践，核心目标是让技术适配人的真实需求。 / HCI/UID is interdisciplinary engineering centered on adapting technology to genuine human needs.

课件依据 / **Source:** chap01_Usability of interactive systems.pdf, p.15-17

Quiz 1 - Q5 (3 pts)

English: Which statement is wrong?

中文: 下列哪一项错误?

A	Progress is more likely from hard work that tunes UID to genuine user needs.	进步更可能来自使 UID 适配真实用户需求的扎实工作。
B	Universal usability and social participation will influence the next phase of interaction.	通用可用性和社会参与将影响下一阶段交互。
C	The ultimate goal of UIs is full automation without involving humans.	UI 的终极目标是完全自动化并排除人。
D	Excellent user experiences can help healthcare, sustainability, resources, and conflict resolution.	优秀体验可支持医疗、可持续发展、资源保护和冲突解决。

答案 / Answer: C

中文讲解: UI 的定义本身就是用户与系统交互的媒介。自动化可以减少某些操作，但“排除人”不是 UID 的终极目标；目标是服务人的需求。

English explanation: A UI is the medium through which people interact with a system. Automation may reduce some operations, but removing people is not the ultimate goal; serving human needs is.

知识点 / Key point:

以人为中心不等于反对自动化；判断标准是自动化是否改善人的目标达成、控制感与安全性。 / Human-centered design does not reject automation; it asks whether automation improves goal achievement, control, and safety.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.12 and p.21

Quiz 1 - Q6 (3 pts)

English: Among the following goals, which is not a usability goal?

中文: 下列哪一项不是课件列出的可用性目标?

A	Usefulness	有用性
B	Universality	通用性
C	Uniqueness	独特性
D	Usability	可用性

答案 / Answer: C

中文讲解: 课件列出的三项目标是 usability、universality、usefulness。Uniqueness 不在其中。

English explanation: The three slide-defined goals are usability, universality, and usefulness. Uniqueness is not one of them.

知识点 / Key point: 目标三联：好用、适用于多样用户、真正有用。 / The three goals are ease/quality of use, coverage of diverse users, and real usefulness.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.22 and p.40

Quiz 1 - Q7 (5 pts)

English: For UIs dedicated to professional environments, which usability measure is normally cared about more than the others? Explain why.

中文: 面向专业环境的 UI 通常更重视哪项可用性度量？说明原因。

A	Time to learn	学习时间
B	Retention over time	长期保持
C	Subjective satisfaction	主观满意度
D	Speed of performance	任务执行速度

答案 / Answer: D

中文讲解: 专业用户可接受较长培训，以换取快速、低错误率的任务执行；高频使用和应急演练可维持记忆，主观满意度相对不是首要指标。

English explanation: Professional users may accept longer training in exchange for fast, low-error task performance. Frequent use and drills support retention, while subjective satisfaction is relatively less central.

知识点 / Key point: 不同场景必须给可用性指标排序；专业环境常优先性能与可靠性。 / Usability measures must be prioritized by context; professional environments often emphasize performance and reliability.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.26-28 and p.35

Quiz 1 - Q8 (3 pts)

English: Which item is not related to individual diversity?

中文: 哪一项不属于个体多样性？

A	Variations in physical abilities	身体能力差异
B	Diverse cognitive abilities	认知能力差异
C	Personality differences	人格差异
D	Cultural differences	文化差异

答案 / Answer: D

中文讲解: 身体、认知和人格差异属于个体差异；文化差异在课件中归入群体/社区多样性。

English explanation: Physical, cognitive, and personality differences are individual diversity. Cultural differences are classified under group/community diversity.

知识点 / Key point: 分类题要先确定分析层级：个人属性，还是群体、文化与社会环境。 / For classification questions, first determine the level of analysis: individual traits or group/cultural context.

课件依据 / Source: chap02_Universal usability.pdf, p.3-12 and p.13-18

Quiz 1 - Q9 (3 pts)

English: Besides time to learn, speed of performance, retention, and subjective satisfaction, what is the remaining usability measure?

中文: 除学习时间、执行速度、长期保持和主观满意度外，第五项可用性度量是什么？

答案 / Answer: Rate of errors by users / 用户错误率

中文讲解: 第五项是用户在完成基准任务时发生的错误数量与类型，并应关注用户如何恢复和纠正错误。

English explanation: The fifth measure is the number and kinds of errors users make on benchmark tasks, including how users recover from and correct those errors.

知识点 / Key point: 不要把“用户错误处理”与“开发者修复软件缺陷”混为一谈。 / Do not confuse user error handling with developers fixing software bugs.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.26-27

Quiz 1 - Q10 (3 pts)

English: Which item does not belong to the usability motivations listed in the slides?

中文: 哪一项不属于课件列出的可用性动机领域？

A	Fully automated industrial manufacturing and agricultural production	全自动工业制造与农业生产
B	Consumer electronics, e-commerce, and social media	消费电子、电子商务和社交媒体
C	Professional environments and sociotechnical systems	专业环境与社会技术系统
D	Exploratory, creative, and collaborative interfaces	探索、创作与协作界面

答案 / Answer: A

中文讲解: 课件列出的动机包括 B、C、D，以及游戏娱乐。A 不是课件中的分类项。

English explanation: The slide categories include B, C, D, and games/entertainment. Option A is not one of the listed categories.

知识点 / Key point: 本题考课件分类表，不是判断某行业是否“需要 UI”。 / This tests the slide taxonomy, not whether an industry could benefit from UI design.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.32-36

Quiz 1 - Q11 (2 pts)

English: Pull-press wells, tri-branch handrails, infrastructure, administration halls, and languages can be regarded as UIs in their respective systems. True or False?

中文: 压水井、三叉扶手、基础设施、行政大厅和语言能否视为各自系统中的用户界面？

答案 / Answer: True / 正确

中文讲解: UI 是用户与系统交互、系统向用户提供服务的媒介，不只存在于计算机。上述对象都承担人与某种系统之间的交互接口。

English explanation: A UI is any medium through which users interact with a system or receive its service; it is not limited to computers. Each example mediates interaction with its respective system.

知识点 / Key point: 把 UI 理解为“交互媒介”，而不是“屏幕上的图形控件”。 / Treat UI as an interaction medium, not merely graphical controls on a screen.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.12

Quiz 1 - Q12 (6 pts)

English: There is no domain where core technology stayed unchanged but UI made a huge difference. True or False? Give an example if false.

中文: 不存在“核心技术未改变，但 UI 极大提升领域影响力”的情况。判断正误；若错误，举例。

答案 / Answer: False / 错误

中文讲解: 课件明确给出相反结论。以个人计算和移动设备为例，底层计算能力的渐进变化并不能单独解释普及；图形界面、触控、直接操纵和社交工具降低了使用门槛，扩大了用户与使用场景。

English explanation: The slides state the opposite. In personal and mobile computing, incremental core-technology changes alone do not explain mass adoption; graphical UI, touch, direct manipulation, and social tools lowered barriers and expanded users and use cases.

知识点 / Key point: 同一核心能力经过不同交互设计，可产生完全不同的可达性、采用率和社会影响。 / The same core capability can produce very different accessibility, adoption, and social impact under different interaction designs.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.13-14

Quiz 1 - Q13 (2 pts)

English: UID experience, empirical evidence, and theories can be organized into three levels of knowledge. True or False?

中文: UID 经验、实证证据和理论可以组织成三个知识层级。判断正误。

答案 / Answer: True / 正确

中文讲解: 三个层级是：指南、原则、理论。

English explanation: The three levels are guidelines, principles, and theories.

知识点 / Key point: 层级从具体到抽象：指南最具体，原则居中，理论最广泛。 / From concrete to abstract: guidelines, principles, and theories.

课件依据 / Source: chap03A_Guidelines.pdf, p.2-3

Quiz 1 - Q14 (2 pts)

English: Among the three levels of UID knowledge, guidelines are at the highest level. True or False?

中文: 在 UID 三层知识结构中, 指南位于最高层。判断正误。

答案 / Answer: False / 错误

中文讲解: 指南是低层、聚焦具体做法的建议; 原则是中层策略; 理论才是高层、广泛适用的框架。

English explanation: Guidelines are low-level focused advice, principles are middle-level strategies, and theories are the high-level broadly applicable frameworks.

知识点 / Key point: “具体可执行”不等于“理论层级高”。 / Concrete actionability does not imply a higher theoretical level.

课件依据 / **Source:** chap03A_Guidelines.pdf, p.2-3

Quiz 1 - Q15 (2 pts)

English: "Standardize task sequences" and "use unique and descriptive headings" are theories rather than guidelines. True or False?

中文: “标准化任务序列”和“使用唯一且描述性的标题”属于理论而非指南。判断正误。

答案 / Answer: False / 错误

中文讲解: 两条内容都在课件“Guidelines - Navigating the UI”中, 是面向具体设计行为的低层建议。

English explanation: Both statements appear under “Guidelines - Navigating the UI” and are low-level advice for concrete design actions.

知识点 / Key point: 识别层级时看其抽象程度与适用范围, 而不是句子是否听起来正式。 / Classify by abstraction and scope, not by how formal the wording sounds.

课件依据 / **Source:** chap03A_Guidelines.pdf, p.5

Quiz 1 - Q16 (2 pts)

English: Individual habits and hobbies belong to group/community diversity. True or False?

中文: 个人习惯和爱好属于群体/社区多样性。判断正误。

答案 / Answer: False / 错误

中文讲解:

习惯、爱好和人格属于个体层面的差异。文化规范、语言、年龄群体与社区实践才主要属于群体/社区多样性。

English explanation: Habits, hobbies, and personality are individual-level differences. Cultural norms, languages, age groups, and community practices mainly belong to group/community diversity.

知识点 / Key point: 同一特征可能受群体影响, 但分类依据是题目要求的主要分析层级。 / A trait may be socially influenced, but classification follows the primary level of analysis requested.

课件依据 / **Source:** chap02_Universal usability.pdf, p.3-18 and p.27

Quiz 1 - Q17 (6 pts)

English: Software backward compatibility can be incorporated into the usability goals to form a larger set of goals requiring tradeoffs. True or False? Give an example.

中文: 软件向后兼容可以纳入可用性目标，形成需要权衡的更大目标集合。判断正误并举例。

答案 / Answer: True / 正确

中文讲解: 例如新版操作系统保留旧文件格式、快捷键和菜单入口，可提高老用户的通用可用性与记忆保持；但兼容层会增加界面复杂度、维护成本和新用户认知负担。设计者需明确保留、迁移或淘汰哪些旧行为。

English explanation: For example, a new OS may preserve old file formats, shortcuts, and menu paths, improving universality and retention for existing users. The compatibility layer can increase UI complexity, maintenance cost, and novice cognitive load, so designers must explicitly decide what to retain, migrate, or retire.

知识点 / Key point: 通用可用性目标不是免费附加项；它会与简洁性、学习成本、性能和维护成本发生冲突。 / Universal-usability goals are not free additions; they can conflict with simplicity, learnability, performance, and maintenance cost.

课件依据 / Source: chap02_Universal usability.pdf, p.26-27; chap01_Usability of interactive systems.pdf, p.28

缺失题目 / Missing questions: 提供的 Quiz1/1.1.jpg 与 Quiz1/1.2.jpg 未包含 Q18、Q19；试卷总分也表明仍缺 4 分内容。本文保留编号空缺。

Quiz 1 - Q20 (4 pts)

English: UIDer A uses pure text prompts; UIDer B provides immersion in a Barbapapa story context. Which better suits children and which better suits adults? Explain.

中文: 方案 A 使用纯文字提示；方案 B 让用户沉浸在孩子熟悉的 Barbapapa 故事情境中。哪种更适合儿童，哪种更适合成人？说明原因。

答案 / Answer: Children: B; Adults: A tends to fit better / 儿童选 B；成人通常更适合 A

中文讲解: 儿童偏好熟悉角色、探索环境、重复与即时反馈，同时抽象能力和注意时长有限，因此 B 更合适。成人通常具备更强的抽象阅读能力，纯文字提示更直接高效；但这只是倾向，仍应按任务和用户测试验证。

English explanation: Children prefer familiar characters, exploration, repetition, and immediate feedback, while having lower abstraction capacity and shorter attention spans, so B is preferable. Adults generally handle abstract text more efficiently, making A more direct; this is a tendency that still requires task-specific user testing.

知识点 / Key point: 面向年龄群体的设计应基于能力与行为特征，不应把偏好刻板化。 / Age-oriented design should be based on capability and behavior, not rigid stereotypes.

课件依据 / Source: chap02_Universal usability.pdf, p.24-25 and p.27

Quiz 1 - Q21 (8 pts)

English: Choose a UID alternative. Time to learn is the priority, subject to performance ≤ 30 min, error rate $\leq 10\%$, retention $\geq 80\%$, and satisfaction $\geq 70/100$.

中文: 在执行时间不超过 30 分钟、错误率不超过 10%、一周保持率不低于 80%、满意度不低于 70/100 的约束下，以学习时间最短为首要目标，选择方案。

答案 / Answer: C

中文讲解: 先做硬约束筛选：A 的错误率 15%，淘汰；B、C、D 均满足约束。再按首要目标比较学习时间：C 为 240 分钟，优于 B 的 360 分钟和 D 的 480 分钟，因此选 C。

English explanation: Apply hard constraints first: A fails with a 15% error rate; B, C, and D remain feasible. Then optimize the stated priority: C has a 240-minute learning time, better than B at 360 and D at 480, so choose C.

知识点 / Key point: 多指标决策的顺序是：先满足门槛，再在可行集中优化首要指标。 / For multi-criteria decisions, filter by constraints first, then optimize the primary objective within the feasible set.

课件依据 / Source: chap01_Usability of interactive systems.pdf, p.26-28

方案	学习时间	执行时间	错误率	保持率	满意度
A	120 min	15 min	15%	80%	75/100
B	360 min	15 min	5%	95%	90/100
C	240 min	25 min	7%	90%	85/100
D	480 min	20 min	5%	97%	90/100

Quiz 2 逐题讲解 / Detailed Solutions

覆盖 / Coverage: mainly Chapter 4 to Chapter 8.

Quiz 2 - Q1 (2 pts)

English: The waterfall model includes the following phases except ____.

中文: 瀑布模型不包括下列哪一阶段？

A	Design and development	设计与开发
B	Testing and evaluation	测试与评估
C	Deployment and maintenance	部署与维护
D	Marketing and commercialization	营销与商业化

答案 / Answer: D

中文讲解:

课件瀑布模型依次包括需求、设计、开发、测试、部署和维护。营销与商业化不是该模型中的工程阶段。

English explanation: The slide waterfall model contains requirements, design, development, testing, deployment, and maintenance. Marketing/commercialization is not one of its engineering phases.

知识点 / Key point: 瀑布模型按阶段顺序推进，主要缺点是反馈来得较晚。 / The waterfall model progresses by sequential phases, with feedback often arriving late.

课件依据 / Source: chap04_Design.pptx, slide 5

Quiz 2 - Q2 (2 pts)

English: Which phases constitute the Four-Phase Iterative Design Process?

中文: 四阶段迭代设计过程由哪些阶段构成？

A	Requirements, large-scale survey, build, field experiment	需求、大规模调查、构建、现场实验
B	Marketing, survey, build, evaluation	营销、调查、构建、评估
C	Requirements analysis, preliminary and detailed design, build and implementation, evaluation	需求分析、初步与详细设计、构建与实现、评估
D	Marketing, design, build, field experiment	营销、设计、构建、现场实验

答案 / Answer: C

中文讲解: 课件给出的四阶段原文即 C，并通过评估反馈回前序阶段形成迭代。

English explanation: The four phases shown in the slides are exactly those in C, with evaluation feeding back into earlier phases.

知识点 / Key point: “迭代”来自反馈闭环，不是把瀑布阶段简单换名。 / Iteration comes from the feedback loop, not merely renaming waterfall phases.

课件依据 / Source: chap04_Design.pptx, slide 7 and slide 13

Quiz 2 - Q3 (2 pts)

English: Which statement is wrong?

中文: 下列哪一项错误?

A	Today's users have backgrounds very different from programmers and engineers.	当代用户背景与程序员、工程师差异很大。
B	Users focus more on professional or recreational needs than on technology itself.	用户更关注职业或娱乐需求，而非技术本身。
C	Effective UIDers start from intended users and facilitate the artifact's function.	有效的 UID 从目标用户出发，并促进其完成系统功能。
D	UIDers should always have a technology-first 'tech brain'.	UID 从业者应始终坚持技术优先的“技术脑”。

答案 / Answer: D

中文讲解: 课件强调以目标用户及其任务为起点。技术是手段，不应凌驾于用户目标之上，因此 D 与用户中心设计相反。

English explanation: The slides emphasize starting from intended users and their tasks. Technology is a means rather than the primary end, so D contradicts user-centered design.

知识点 / Key point: 高技术含量不自动等于高可用性或高价值。 / Technical sophistication does not automatically produce usability or value.

课件依据 / Source: chap04_Design.pptx, slide 4

Quiz 2 - Q4 (2 pts)

English: Which pair is not a correct example of a UI metaphor?

中文: 哪一组不是正确的 UI 隐喻对应关系?

A	Desktop GUI - physical desk	桌面 GUI - 实体书桌
B	Web portal/page - magazine	网页/门户 - 杂志页面
C	Shopping cart - file hierarchies	购物车 - 文件层级
D	Toolbar/menu bar - workbench	工具栏/菜单栏 - 工作台

答案 / Answer: C

中文讲解: 购物车模拟线下购物；文件层级应对应文件夹/树形结构。C 把两个不同隐喻错误配对。

English explanation: A shopping cart mimics in-store shopping, while file hierarchies are represented by folders/tree views. C mismatches two distinct metaphors.

知识点 / Key point: 隐喻由熟悉的源域映射到数字目标域，映射关系必须语义一致。 / A metaphor maps a familiar source domain to a digital target domain, and the semantic correspondence must be coherent.

课件依据 / Source: chap07_Direct Manipulation.pdf, p.3

Quiz 2 - Q5 (2 pts)

English: Direct-manipulation strength ranges from weak to medium, strong, and ____.

中文: 直接操纵隐喻的强度从弱、中、强，继续到哪一级？

A	Useful	有用
B	Universal	通用
C	Subjective	主观
D	Immerse	沉浸

答案 / Answer: D

中文讲解: 课件连续谱为 weak - medium - strong - immerse/immersive。VR 与 AR 被列在沉浸端。

English explanation: The slide continuum is weak - medium - strong - immerse/immersive, with VR and AR at the immersive end.

知识点 / Key point: 平移距离越小，用户感到自己越直接地作用于对象。 / The smaller the translational distance, the more directly users feel they are acting on the object.

课件依据 / **Source:** chap07_Direct Manipulation.pdf, p.4

Quiz 2 - Q6 (2 pts)

English: Which is not an example of direct manipulation?

中文: 哪一项不是直接操纵的例子？

A	Geographical systems	地理系统
B	Binary and grid menus	二元菜单和网格菜单
C	Computer-Aided Design	计算机辅助设计
D	Video games and digital twins	电子游戏与数字孪生

答案 / Answer: B

中文讲解: 地理系统、游戏、CAD 和数字孪生都在直接操纵示例中。二元菜单和网格菜单属于“通过选择进行导航”。

English explanation: Geographical systems, games, CAD, and digital twins are listed as direct-manipulation examples. Binary and grid menus belong to navigation by selection.

知识点 / Key point: 判断交互风格时看核心动作：直接作用于对象，还是从显式选项中选择。 / Classify by the core action: directly manipulating an object or selecting from explicit choices.

课件依据 / **Source:** chap07_Direct Manipulation.pdf, p.6-8; chap08_Navigation+FFI+DB.pdf, p.3

Quiz 2 - Q7 (2 pts)

English: Which statement is wrong?

中文: 下列哪一项错误?

A	VR generates synthetic images that simulate real or imagined reality.	VR 生成模拟现实或想象世界的合成图像。
B	AR augments captured real-world images with virtual elements.	AR 在真实世界图像上叠加虚拟元素。
C	Navigation by selection usually gives higher subjective satisfaction than direct manipulation.	选择式导航通常比直接操纵带来更高主观满意度。
D	Navigation uses rapid skimming, recognition, and selection.	导航利用快速浏览、识别和选择。

答案 / Answer: C

中文讲解: 课件把高主观满意度列为直接操纵的优势，并未声称选择式导航通常更高。A、B、D 均与课件定义一致。

English explanation: High subjective satisfaction is listed as an advantage of direct manipulation; the slides do not claim navigation by selection is usually higher. A, B, and D match the slide definitions.

知识点 / Key point: 直接操纵的优势来自可见对象、快速反馈、可逆动作与较低语法负担。 / Direct manipulation benefits from visible objects, rapid feedback, reversibility, and reduced syntax burden.

课件依据 / **Source:** chap07_Direct Manipulation.pdf, p.5 and p.12-15; chap08_Navigation+FFI+DB.pdf, p.2

Quiz 2 - Q8 (2 pts)

English: Vivid task concepts, easy learning, easy retention, and fewer errors are advantages of ____.

中文: 生动呈现任务概念、易学、易保持且更少出错，属于哪种交互方式的优势?

A	Navigation by selection	选择式导航
B	Direct manipulation	直接操纵
C	Form fill-in	表单填写
D	Dialog boxes	对话框

答案 / Answer: B

中文讲解: 这些表述逐项对应课件列出的直接操纵优点。

English explanation: These phrases directly match the advantages of direct manipulation listed in the slides.

知识点 / Key point: 直接操纵通过可见、连续、可逆的动作降低学习和错误成本。 / Visible, continuous, reversible actions reduce learning and error costs.

课件依据 / **Source:** chap07_Direct Manipulation.pdf, p.5

Quiz 2 - Q9 (2 pts)

English: The demonstrated seeing-through effect is an effect of ____.

中文: 题图展示的“透视/看穿”效果属于哪种技术?

A	Image thresholding	图像阈值化
B	Image resolution enhancement	图像分辨率增强
C	Augmented reality	增强现实
D	Virtual reality	虚拟现实

答案 / Answer: C

中文讲解: 图中保留真实场景，同时叠加不可直接看见的结构信息，属于在真实图像上增加虚拟元素的 AR，而非完全合成环境的 VR。

English explanation: The image retains the real scene while overlaying otherwise invisible structural information. This is AR, which augments real imagery, not VR, which creates a synthetic environment.

知识点 / Key point: 区分 AR/VR 的最直接问题：是否以真实世界图像为底层。 / The simplest AR/VR test is whether real-world imagery remains the base layer.

课件依据 / Source: chap07_Direct Manipulation.pdf, p.12-15

Quiz 2 - Q10 (2 pts)

English: Which statement about evaluation is wrong?

中文: 关于评估，哪一项错误?

A	Evaluation is only one part of a larger system and resources are limited.	评估只是更大系统的一部分，资源始终有限。
B	It is not a big deal for UIers to recognize that extensive testing and evaluation are necessary.	认识到广泛测试和评估的必要性并不重要。
C	Careful choices are required to create a balanced evaluation.	必须谨慎选择方法以形成平衡评估。
D	Many factors influence when, where, and how evaluation is done.	许多因素会影响评估的时间、地点和方式。

答案 / Answer: B

中文讲解: 承认广泛测试与评估的必要性需要专业判断和谦逊，绝不是“不重要”。其余三项都体现了受约束的系统性评估观。

English explanation: Recognizing the need for extensive testing and evaluation requires professional judgment and humility; it is not trivial. The other statements reflect a constrained, systems-oriented view of evaluation.

知识点 / Key point: 评估不是无限测试，而是在风险、成本、阶段和证据质量之间做平衡。 / Evaluation is not unlimited testing; it balances risk, cost, lifecycle stage, and evidence quality.

课件依据 / Source: Chapter 5/6 evaluation slide deck is not present in the supplied GUI directory; answer audited against the quiz wording and standard HCI terminology.

说明 / Note: 课件缺失说明：目录未提供第5/6章评估课件。

Quiz 2 - Q11 (2 pts)

English: Factors influencing evaluation include the following except ____.

中文: 下列哪一项不是影响评估安排的合理因素?

A	Novelty and criticality of the UI	UI 的新颖性与关键性
B	Number of expected users	预期用户数量
C	Environment where the UI is fabricated	UI 被制造的环境
D	Product cost and testing budget	产品成本与测试预算

答案 / Answer: C

中文讲解: 新颖性、关键性、用户规模、产品成本和测试预算都会影响评估投入。制造地点本身不决定用户评估；真正相关的是预期使用环境。

English explanation: Novelty, criticality, user scale, product cost, and testing budget affect evaluation effort. The fabrication environment itself does not determine user evaluation; the intended use environment does.

知识点 / Key point: 不要混淆开发/制造环境与实际使用情境。 / Do not confuse the development/fabrication environment with the real context of use.

课件依据 / Source: Chapter 5/6 evaluation slide deck is not present in the supplied GUI directory; answer audited against the quiz wording and standard HCI terminology.

说明 / Note: 课件缺失说明：目录未提供第5/6章评估课件。

Quiz 2 - Q12 (2 pts)

English: Which is not a principle of direct manipulation?

中文: 哪一项不是直接操纵原则?

A	Use two-factor authentication for crucial users	为关键用户使用双因素认证
B	Continuous representation with meaningful visual metaphors	以有意义的视觉隐喻持续表示对象和动作
C	Physical actions or labeled objects instead of complex syntax	用物理动作或有标签对象替代复杂语法
D	Rapid, incremental, reversible actions with immediately visible effects	快速、渐进、可逆且效果立即可见的动作

答案 / Answer: A

中文讲解: B、C、D 是课件原列的三项原则。双因素认证属于安全机制，不是直接操纵原则。

English explanation: B, C, and D are the three principles listed in the slides. Two-factor authentication is a security mechanism, not a direct-manipulation principle.

知识点 / Key point: 同一产品可同时使用安全机制和直接操纵，但二者属于不同设计问题。 / A product may use both security mechanisms and direct manipulation, but they solve different design problems.

课件依据 / Source: chap07_Direct Manipulation.pdf, p.2

Quiz 2 - Q13 (2 pts)

English: Which statement is not related to handling uncertainty in evaluation?

中文: 哪一项与处理评估中的不确定性无关?

A	Uncertainty remains, so monitoring and repair must continue through the UI lifecycle.	不确定性会残留，因此需在生命周期内持续监测和修复。
B	A release decision must eventually be made even while problems may remain.	即使仍可能发现问题，也必须在某时点作出发布决策。
C	Project success depends on precise shared understanding among designers, developers, and users.	项目成功取决于设计者、开发者和用户之间准确完整的共同理解。
D	Life-critical UIs require evaluation under stress and partial equipment failure.	生命关键 UI 需评估压力情境和部分设备失效。

答案 / Answer: C

中文讲解: C 讨论需求沟通与共同理解，属于项目协作问题；A、B、D 都直接讨论如何面对评估无法消除的风险与不确定性。

English explanation: C concerns requirements communication and shared understanding, a project-collaboration issue. A, B, and D directly address residual risk and uncertainty in evaluation.

知识点 / Key point: 不确定性处理包括持续监测、停止准则和极端场景测试。 / Handling uncertainty includes continued monitoring, stopping criteria, and extreme-scenario testing.

课件依据 / Source: Chapter 5/6 evaluation slide deck is not present in the supplied GUI directory; answer audited against the quiz wording and standard HCI terminology.

说明 / Note: 课件缺失说明：目录未提供第5/6章评估课件。

Quiz 2 - Q14 (2 pts)

English: The spectrum of usability testing includes the following except ____.

中文: 可用性测试谱系不包括哪一项?

A	Expert reviews	专家评审
B	Paper mockups and prototyping	纸面模型与原型
C	A/B (Alice/Bob) Testing	A/B (Alice/Bob) 测试
D	Can-You-Break-This testing	破坏性尝试测试

答案 / Answer: C

中文讲解: A/B testing 中的 A/B 指两个版本或条件，不是 Alice/Bob。题目故意用错误展开制造干扰。

English explanation: In A/B testing, A and B denote two variants or conditions, not Alice and Bob. The option is a deliberately false expansion.

知识点 / Key point: 术语题应掌握定义，不要仅凭熟悉的缩写形式猜测。 / For terminology questions, rely on the definition rather than a familiar-looking expansion.

课件依据 / Source: Chapter 5/6 evaluation slide deck is not present in the supplied GUI directory; answer audited against the quiz wording and standard HCI terminology.

说明 / Note: 课件缺失说明：目录未提供第5/6章评估课件。

Quiz 2 - Q15 (2 pts)

English: Which statement is wrong about acceptance testing?

中文: 关于验收测试，哪一项错误？

A	Pilot testing refines benchmark tasks, materials, and procedures.	先导测试用于完善基准任务、材料与流程。
B	Acceptance testing may be adversarial and use outside organizations for neutrality.	验收测试可能具有对抗性，并由外部机构保证中立。
C	Its central goal is not verifying requirements/contracts but detecting flaws.	核心目标不是验证需求/合同，而是发现缺陷。
D	Field or beta testing may follow successful acceptance testing.	验收通过后仍可能进行现场或大规模 Beta 测试。

答案 / Answer: C

中文讲解: 验收测试的核心正是确认系统是否满足事先约定的需求或合同标准；发现缺陷可以发生，但不是替代该核心目标。

English explanation: The central goal of acceptance testing is precisely to verify conformance to agreed requirements or contractual criteria. Defect discovery may occur but does not replace that goal.

知识点 / Key point: 可用性测试关注使用质量；验收测试关注是否达到预先定义的接受标准。 / Usability testing examines quality of use; acceptance testing checks predefined acceptance criteria.

课件依据 / Source: Chapter 5/6 evaluation slide deck is not present in the supplied GUI directory; answer audited against the quiz wording and standard HCI terminology.

说明 / Note: 课件缺失说明：目录未提供第5/6章评估课件。

Quiz 2 - Q16 (10 pts)

English: Construct a statistical example of Simpson's paradox, explain it, and state the scientific lessons.

中文: 构造一个统计例子说明辛普森悖论，解释原因，并总结科学启示。

答案 / Answer: See the stratified example below / 见下方分层示例

中文讲解: 设比较 UI A 与 UI B 的任务成功率。新手层：A 为 $30/100=30\%$ ，B 为 $1/10=10\%$ ；专家层：A 为 $9/10=90\%$ ，B 为 $80/100=80\%$ 。A 在两个层内都更好，但汇总后 A 为 $39/110=35.5\%$ ，B 为 $81/110=73.6\%$ ，结论反转。原因是 A 的样本大多是新手，B 的样本大多是专家，用户体验是混杂变量。

English explanation: Compare task success for UI A and UI B. Novices: A is $30/100=30\%$, B is $1/10=10\%$. Experts: A is $9/10=90\%$, B is $80/100=80\%$. A wins in both strata, yet overall A is $39/110=35.5\%$ and B is $81/110=73.6\%$, reversing the conclusion. A was tested mostly on novices and B mostly on experts, so experience level is a confounder.

知识点 / Key point: 科学启示：汇总相关不等于因果；必须检查分层和样本构成；预先识别混杂变量；随机化或匹配；同时报告总体与分层结果；用领域知识决定是否合并数据。 / Scientific lessons: aggregate association is not causation; inspect strata and sample composition; identify confounders in advance; randomize or match; report both aggregate and stratified results; use domain knowledge to decide whether pooling is valid.

课件依据 / Source: Chapter 5/6 evaluation slide deck is not present in the supplied GUI directory; numerical example constructed for this explanation.

说明 / Note: 课件缺失说明：目录未提供包含辛普森悖论的评估课件。

用户层 / Stratum	UI A	UI B	层内结论 / Within-stratum
新手 / Novice	30/100 = 30%	1/10 = 10%	A > B
专家 / Expert	9/10 = 90%	80/100 = 80%	A > B
汇总 / Overall	39/110 = 35.5%	81/110 = 73.6%	B > A (反转)

核心复习清单 / Final Review Checklist

主题 / Topic	必须掌握 / Must know
可用性 / Usability	三项目标；五项度量；场景化权衡
通用可用性 / Universal usability	个体与群体差异；儿童；软硬件兼容
知识层级 / Knowledge levels	指南-原则-理论，从具体到抽象
设计过程 / Design process	瀑布模型；四阶段迭代与反馈闭环
直接操纵 / Direct manipulation	三原则；隐喻；强度连续谱；优缺点
AR/VR 与导航	真实底图与合成环境；对象操纵与菜单选择
评估 / Evaluation	评估约束、验收标准、分层分析与混杂变量

考试作答 / Exam writing: 判断题和选择题不仅写结论；解释题应给出定义、排除理由、适用条件和一个具体例子。涉及多指标时先写约束筛选，再写优化目标。

主要资料 / Main materials: Quiz1/1.1.jpg, Quiz1/1.2.jpg, Quiz2/1.jpg-4.jpg, and the chapter lecture files in GUI开发.